

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275868

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Ryan; Edwin

INFUSION DEVICE AND METHOD

Abstract

An ophthalmic infusion support and associated methods are shown. Example infusion support devices include a base unit, a fixation device coupled to the bottom surface of the base unit, and an infusion tube holding device configured to hold a loop of an infusion tube an angle projecting upward from the patient.

Inventors: Ryan; Edwin (St. Paul, MN)

Applicant: Ryan; Edwin (St. Paul, MN)

Family ID: 55748123

Appl. No.: 19/084096

Filed: March 19, 2025

Related U.S. Application Data

parent US continuation 18238348 20230825 parent-grant-document US 12263120 child US 19084096

parent US continuation 14885274 20151016 ABANDONED child US 18238348

us-provisional-application US 62154462 20150429

us-provisional-application US 62065428 20141017

Publication Classification

Int. Cl.: A61F9/00 (20060101)

U.S. Cl.:

CPC A61F9/0026 (20130101); A61F2009/0052 (20130101)

Background/Summary

CLAIM OF PRIORITY [0001] This application is a continuation of and claims the benefit of priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 18/238,348, filed on Aug. 25, 2023, which is a continuation of and claims the benefit of priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 14/885,274, filed on Oct. 16, 2015, which claims the benefit priority to U.S. Provisional Patent Application No. 62/065,428, filed Oct. 17, 2014, and to U.S. Provisional Patent Application No. 62/154,462, filed Apr. 29, 2015 which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] This invention relates to devices and methods for ophthalmological procedures, such as an infusion procedure.

BACKGROUND

[0003] A number of ophthalmological procedures require the use of an infusion line. Infusion lines have weight, and in some cases, may tend to flex as they hang from their coupling location on a patient's eye. It is desirable to avoid kinking and/or twisting of the infusion line, which may cause problems such as restricting fluid flow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 shows an ophthalmic infusion support according to an embodiment of the invention.

[0005] FIG. 2 shows another ophthalmic infusion support according to an embodiment of the invention.

[0006] FIG. 3 shows one example of an ophthalmic infusion support in use according to an embodiment of the invention.

[0007] FIG. 4A shows another example of an ophthalmic infusion support according to an embodiment of the invention.

[0008] FIG. 4B shows an isometric view of an ophthalmic infusion support according to an embodiment of the invention.

[0009] FIG. 5 shows another ophthalmic infusion support according to an embodiment of the invention.

[0010] FIG. 6 shows another ophthalmic infusion support according to an embodiment of the invention.

[0011] FIG. 7 shows another ophthalmic infusion support according to an embodiment of the invention.

[0012] FIG. 8 shows another ophthalmic infusion support according to an embodiment of the invention.

[0013] FIG. 9 shows an example method of using an ophthalmic infusion support according to an embodiment of the invention.

[0014] FIG. 10 shows another ophthalmic infusion support according to an embodiment of the invention.

[0015] FIG. 11 shows another ophthalmic infusion support according to an embodiment of the invention.

DETAILED DESCRIPTION

[0016] In the following detailed description, reference is made to the accompanying drawings which form a part hereof, and in which is shown, by way of illustration, specific embodiments in which the invention may be practiced. In the drawings, like numerals describe substantially similar

components throughout the several views. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments may be utilized and structural, or logical changes, etc. may be made without departing from the scope of the present invention.

[0017] FIG. 1 shows an ophthalmic infusion support **100** according to one example of the invention. The ophthalmic infusion support **100** includes a base unit **102** having a bottom surface **104** and a top surface **106**. In one example the base unit **102** is substantially flat in one dimension (for example in the plane of the page of FIG. 1). In such an example the base unit **102** may be cut from a sheet of material, such as a closed or open cell polymer foam sheet. In one example, a resilient material such as foamed polymer sheet provides a desirable level of compliance that may allow a user to conform the ophthalmic infusion support **100** to a patient for improved fixation as discussed below. In other examples, the base unit **102** may be molded from a compliant polymer, or rigid polymer. Other materials such as metals are also possible, but may add unwanted cost to the ophthalmic infusion support **100**.

[0018] In the example of FIG. 1, an adhesive **120** is coupled to the bottom surface **104**. The adhesive serves as a fixation device, and may be used to attach to a patient. In one example the adhesive such as a pressure sensitive adhesive is used. In the example of FIG. 1, a peel off backing **122** is shown that may be removed to expose the adhesive **120** for attaching to a patient.

[0019] In one example, the adhesive **120** is used to attach directly to a patient, such as adhering to the patient's skin in a region adjacent to an eye, as will be discussed in more detail below. In one example, a fixation device, such as the adhesive **120** is used to attach indirectly to a patient, such as attaching to a surgical drape that is placed on the patient. Both direct fixation, and indirect fixation methods are encompassed in the present disclosure. Any number of suitable adhesives may be used as a fixation device, as discussed above. Other fixation devices include, but are not limited to, mechanical fixation, such as spring clips, pins, etc. Mechanical fixation device examples will primarily be used for indirect fixation to an intermediate element, such as a surgical drape.

[0020] The ophthalmic infusion support **100** of FIG. 1 further shows an infusion tube holding device **110** coupled to the top surface **106** of the base unit **102**. In the example of FIG. 1, the infusion tube holding device **110** includes an adhesive **110**. The adhesive **110** may be the same or similar adhesive to the adhesive **120**. In one example, a first adhesive **110** is chosen to securely hold an infusion tube, while a second, different adhesive **120** is chosen to adhere to the patient, either directly or indirectly.

[0021] In one example an adhesive such as a pressure sensitive adhesive is used for the infusion tube holding device **110**. Similar to the adhesive **120**, in the example of FIG. 1, a peel off backing **112** is shown that may be removed to expose the adhesive **110** for attaching to an infusion tube.

[0022] Although an adhesive is shown as the infusion tube holding device **110**, the invention is not so limited. Other infusion tube holding devices may include mechanical holding devices, such as clips, clamps, fasteners, zip ties, etc. may be used as an infusion tube holding device **110**.

[0023] FIG. 2 shows an ophthalmic infusion support **200** according to an example of the invention. The ophthalmic infusion support **200** includes a base unit **202** having a bottom surface **204** and a top surface **206**. Similar to the ophthalmic infusion support **100**, in one example the base unit **202** is substantially flat in one dimension (for example in the plane of the page of FIG. 1). As described above, materials for the base unit **202** may include, but are not limited to, polymers, foamed polymers, metals, composites, etc.

[0024] Similar to the example of FIG. 1, an adhesive **220** is coupled to the bottom surface **204**. The adhesive serves as a fixation device, and may be used to attach to a patient, directly or indirectly as discussed above. In one example the adhesive such as a pressure sensitive adhesive is used. In the example of FIG. 2, a peel off backing **222** is shown that may be removed to expose the adhesive **220** for attaching to a patient. In the example of FIG. 2, the bottom surface **204** is anatomically curved to better conform and adhere to a patient, either directly or indirectly.

[0025] Similar to the example of FIG. 1, in one example an infusion tube holding device **210** is coupled to the top surface **206** of the base unit **202**. In the example of FIG. 2, the infusion tube holding device **201** includes an adhesive **210**. In one example an adhesive such as a pressure sensitive adhesive is used for the infusion tube holding device **210**. Similar to the adhesive **220**, a peel off backing **212** is shown that may be removed to expose the adhesive **210** for attaching to an infusion tube.

[0026] FIG. 3 shows an example of an ophthalmic infusion support **300**, similar to infusion supports **100** and **200** discussed above. The ophthalmic infusion support **300** is shown attached to a patient adjacent to the patient's eye. An infusion tube **310** is shown coupled to the ophthalmic infusion support **300**. The infusion tube **310** includes a supply end **312**, and an outlet end **314**. A loop **316** of the infusion tube **310** is shown adjacent to the patient's eye. The loop projects upward from the patient as a result of being coupled to the ophthalmic infusion support **300**.

[0027] The loop **316**, and the upward projection provide comfort to the patient by allowing the infusion tube to move slightly without any excessive forces to the patient's eye. Example infusion support devices, such as infusion supports **100**, **200**, **300** and other examples as described in the present disclosure, help resist the weight of the tubing that may tend to pull a tip of the infusion line into unwanted orientations. The loop **316**, and the upward projection provide an infusion supply that is directed toward a desired portion of the eye, with unobstructed flow, and is much less likely to kink, twist, or otherwise become obstructed during a procedure.

[0028] FIG. 4A further illustrates how examples of ophthalmic infusion supports provide a secure loop with an upward projection. In the example of FIG. 4, the infusion tube **310** is attached to the ophthalmic infusion support **300** along a top surface. In one example, the infusion tube **310** is attached using an adhesive as described in FIGS. 1 and 2. Other examples of attachment without adhesives are described in additional examples below, including FIG. 4B. The infusion tube **310** of FIG. 4, includes a loop **316** that projects upward from the patient as depicted by surface **304**. The loop **316** projects upward at an angle **320** from the surface **304**. In one example, the upward projection of the loop **316** provides a downward projection of the supply end **314** of the infusion tube **310**. Such a configuration provides a reliable, substantially fixed angle of supply to an eye of the patient without unwanted misdirection of infusion flow, obstruction of infusion flow, kinking, twisting, etc. of the infusion tube **310**.

[0029] In the example of FIG. 4A, the upward projection of the loop **316** is a direct result of aligning the infusion tube **310** along an angled top surface of the ophthalmic infusion support **300**, however, the invention is not so limited. Other configurations may use other types of infusion tube holding devices to provide an upward projection of a loop. Selected other examples are discussed in more detail below.

[0030] FIG. 4B shows an example of an ophthalmic infusion support **400**, including a base unit **402** having a bottom surface **404** and a top surface **406**. Similar to other example ophthalmic infusion supports (**100**, **200**, **300**) in one example the base unit **402** is substantially flat in one dimension. As described above, materials for the base unit **402** may include, but are not limited to, polymers, foamed polymers, metals, composites, etc.

[0031] In one example, the bottom surface **404** includes an adhesive coupled to the bottom surface **404**. As in other examples described, the adhesive may serve as a fixation device, and may be used to attach to a patient. In one example an adhesive such as a pressure sensitive adhesive is used. A peel off backing for an adhesive may also be used in selected embodiments. As in other embodiments described, the bottom surface **404** may be flat, or curved. As discussed above, in alternate examples, other fixation devices that may be used in place of adhesive **520** include, but are not limited to, mechanical fixation, such as spring clips, pins, etc. Mechanical fixation device examples will primarily be used for indirect fixation to an intermediate element, such as a surgical drape.

[0032] In the Example of FIG. 4B, a pair of infusion tube holding devices are shown, including a

first infusion tube holding device **410** and a second infusion tube holding device **412**. In the example shown, the infusion tube holding devices **410**, **412** include resilient clips, although the invention is not so limited. In the resilient clip example of FIG. **4B**, arms of the infusion tube holding devices **410**, **412** are flexible and can deform to allow insertion of an infusion tube (not shown) into the resilient clips. The pair of infusion tube holding devices **410**, **412** of the ophthalmic infusion support **400** are configured to hold a portion of an infusion tube in a loop that projects upward from a patient, as shown in FIG. **4A** above.

[0033] FIG. **5** shows an example of an ophthalmic infusion support **500**, including a base unit **502** having a bottom surface **504** and a top surface **506**. Similar to other example ophthalmic infusion supports (**100**, **200**, **300**) in one example the base unit **502** is substantially flat in one dimension (for example in the plane of the page of FIG. **5**). As described above, materials for the base unit **502** may include, but are not limited to, polymers, foamed polymers, metals, composites, etc.

[0034] The example of FIG. **5** shows a pair of infusion tube holding devices, including a first infusion tube holding device **510** and a second infusion tube holding device **512**. In the example shown, the infusion tube holding devices **510**, **512** include resilient clips, although the invention is not so limited. In the resilient clip example of FIG. **5**, arms **511** of the infusion tube holding devices **510**, **512** are flexible and can deform to allow insertion of an infusion tube (not shown) into the resilient clips. The pair of infusion tube holding devices **510**, **512** of the ophthalmic infusion support **500** are configured to hold a portion of an infusion tube in a loop that projects upward from a patient, as shown in more detail in FIGS. **7** and **8** below.

[0035] Similar to other example ophthalmic infusion supports (**100**, **200**, **300**), an adhesive **520** is coupled to the bottom surface **504** of the base unit **502**. The adhesive **520** serves as a fixation device, and may be used to attach to a patient, directly or indirectly as discussed above. In one example the adhesive such as a pressure sensitive adhesive is used. In the example of FIG. **5**, a peel off backing **522** is shown that may be removed to expose the adhesive **520** for attaching to a patient. As discussed in other examples, the bottom surface **504** may be flat as shown in FIG. **5**, or optionally anatomically curved as shown in FIG. **2** to better conform and adhere to a patient, either directly or indirectly.

[0036] As discussed above, in alternate examples, other fixation devices that may be used in place of adhesive **520** include, but are not limited to, mechanical fixation, such as spring clips, pins, etc. Mechanical fixation device examples will primarily be used for indirect fixation to an intermediate element, such as a surgical drape.

[0037] FIG. **6** shows an example of an ophthalmic infusion support **600**, including a base unit **602** having a bottom surface **604** and a top surface **606**. Similar to other example ophthalmic infusion supports (**100**, **200**, **300**, **500**) in one example the base unit **602** is substantially flat in one dimension although other configurations are within the scope of the invention. As described above, materials for the base unit **602** may include, but are not limited to, polymers, foamed polymers, metals, composites, etc.

[0038] The example of FIG. **6** shows a pair of infusion tube holding devices, including a first infusion tube holding device **610** and a second infusion tube holding device **612**. In the example shown, the infusion tube holding devices **610**, **612** include resilient clips, although the invention is not so limited.

[0039] In the resilient clip example of FIG. **6**, the infusion tube holding devices **610**, **612** are integrally molded or cut from the base unit **602**. Similar to the example of FIG. **5**, the infusion tube holding devices **610**, **612** are flexible and can deform to allow insertion of an infusion tube (not shown) into the resilient clips. In one example by integrally molding or cutting the infusion tube holding devices **610**, **612** from the base unit **602**, manufacturing cost and materials cost savings are provided. The pair of infusion tube holding devices **610**, **612** of the ophthalmic infusion support **600** are configured to hold a portion of an infusion tube in a loop that projects upward from a patient, as shown in more detail in FIGS. **7** and **8** below.

[0040] Similar to other example ophthalmic infusion supports (**100**, **200**, **300**, **500**), an adhesive **620** is coupled to the bottom surface **604** of the base unit **602**. The adhesive **620** serves as a fixation device, and may be used to attach to a patient, directly or indirectly as discussed above. In one example the adhesive such as a pressure sensitive adhesive is used. In the example of FIG. **6**, a peel off backing **622** is shown that may be removed to expose the adhesive **620** for attaching to a patient. As discussed in other examples, the bottom surface **604** may be flat as shown in FIG. **6**, or optionally anatomically curved as shown in FIG. **2** to better conform and adhere to a patient, either directly or indirectly.

[0041] As discussed above, in alternate examples, other fixation devices that may be used in place of adhesive **620** include, but are not limited to, mechanical fixation, such as spring clips, pins, etc. Mechanical fixation device examples will primarily be used for indirect fixation to an intermediate element, such as a surgical drape.

[0042] FIGS. **7** and **8** further illustrate how examples of ophthalmic infusion supports shown in FIGS. **5** and **6** provide a secure loop with an upward projection. In the example of FIG. **7**, an infusion tube **710** is attached to an ophthalmic infusion support **700**. In one example, the infusion tube **710** is attached using a pair of infusion tube holding devices **701**, **703**, two possible examples of which are shown in FIGS. **5** and **6**.

[0043] The ophthalmic infusion support **700** is shown attached to a patient adjacent to the patient's eye **730**. The infusion tube **710** includes a supply end **712**, and an outlet end **714**. A loop **716** of the infusion tube **710** is shown adjacent to the patient's eye. The loop projects upward from the patient, as further illustrated in FIG. **8**, as a result of being coupled to the ophthalmic infusion support **700**.

[0044] As shown in FIG. **8**, the loop **716** projects upward from the patient as depicted by surface **704**. The loop **716** projects upward at an angle **720** from the surface **704**. In one example, the upward projection of the loop **716** provides a downward projection of the supply end **714** of the infusion tube **710**. Such a configuration provides a reliable, substantially fixed angle of supply to an eye of the patient without unwanted kinking, twisting, etc. of the infusion tube **710**.

[0045] In the example of FIG. **8**, the upward projection of the loop **716** is provided by the arrangement and orientation of the pair of infusion tube holding devices **701**, **703**. In one example, both the arrangement and orientation of the pair of infusion tube holding devices **701**, **703**, and the upward angle of the top surface of the base unit **702** provide the upward projection of the loop **716**. The loop **716**, and the upward projection provide comfort to the patient by allowing the infusion tube to move slightly without any excessive forces to the patient's eye. The loop **716**, and the upward projection further provide a reliable, substantially fixed angle of supply to an eye of the patient without unwanted misdirection of infusion flow, obstruction of infusion flow, kinking, twisting, etc. of the infusion tube during a procedure.

[0046] FIG. **9** shows a method of using an ophthalmic infusion support, such as one or more of the ophthalmic infusion supports described in examples above. In operation **902**, a cannula is inserted into a patient's eye. In operation **904**, an infusion tube is coupled to the cannula. In operation **906**, a loop is formed with the infusion tube that projects upward from the patient at an angle. In operation **908**, an ophthalmic infusion support is fixed to the patient. In operation **910**, the infusion tube is coupled to an ophthalmic infusion support that holds the infusion tube at the angle.

[0047] FIG. **10** shows an example of an ophthalmic infusion support **1000**, including a base unit **1002** having a bottom surface **1004** and a top surface **1006**. As described above, materials for the base unit **1002** may include, but are not limited to, polymers, foamed polymers, metals, composites, etc.

[0048] In one example, the bottom surface **1004** includes an adhesive coupled to the bottom surface **1004**. As in other examples described, the adhesive may serve as a fixation device, and may be used to attach to a patient. In one example an adhesive such as a pressure sensitive adhesive is used. A peel off backing for an adhesive may also be used in selected embodiments. As in other embodiments described, the bottom surface **1004** may be flat, or curved. As discussed above, in

alternate examples, other fixation devices that may be used in place of adhesive include, but are not limited to, mechanical fixation, such as spring clips, pins, etc.

[0049] In the Example of FIG. **10**, a single, elongated infusion tube holding device **1008** is shown. In one example, the infusion tube holding device **1008** is integrally molded or cut from the base unit **1002**. Similar to the example of FIG. **5**, the infusion tube holding device **1008** is flexible and can deform to allow insertion of an infusion tube (not shown). In one example by integrally molding or cutting the infusion tube holding device **1008** from the base unit **1002**, manufacturing cost and materials cost savings are provided.

[0050] FIG. **11** shows an example of an ophthalmic infusion support **1100**, including a base unit **1102** having a bottom surface **1104**. In one example, the bottom surface **1104** includes an adhesive coupled to the bottom surface **1104**. As in other examples described, the adhesive may serve as a fixation device, and may be used to attach to a patient. In one example an adhesive such as a pressure sensitive adhesive is used. A peel off backing for an adhesive may also be used in selected embodiments. As in other embodiments described, the bottom surface **1104** may be flat, or curved. As discussed above, in alternate examples, other fixation devices that may be used in place of adhesive include, but are not limited to, mechanical fixation, such as spring clips, pins, etc.

[0051] The ophthalmic infusion support **1100** example of FIG. **11** shows a first infusion tube holding device **1110** and a second infusion tube holding device **1112**. In one example, the ophthalmic infusion support **1100** is wholly or partially formed from a resilient material, such as a polymer material, or flexible metal. In one resilient material example, a top portion **1106** may be tilted slightly to one side or the other in order to facilitate insertion of an infusion tube into the first or second infusion tube holding device **1110**, **1112**. The resilient nature of the material will then serve as a clamping force to hold the infusion tube in place within the chosen infusion tube holding device. One advantage of two infusion tube holding devices **1110**, **1112** includes a greater number of options to hold the infusion tube in place without obstructing view of the eye for the procedure.

[0052] Although in one example, the entire ophthalmic infusion support **1100** is formed from a resilient material, although the invention is not so limited. In one example, only a portion, such as a connecting region **1108** is formed from a resilient material. In other examples, the ophthalmic infusion support **1100** is formed from a rigid material, and a resilient nature of the infusion tube is used to hold the tube within either one of the first or second infusion tube holding devices **1110**, **1112**.

[0053] In all examples shown, the infusion tube holding devices provide an important feature of permitting a user to adjust an amount of infusion tube that extends beyond an end of an ophthalmic infusion support toward the patient. This allows a user to adjust a correct length of infusion tube to provide an optimal length of tubing to form a loop that projects upward from the patient. As discussed above, in one example, the upward projection of a loop provides a downward projection of a supply end of the infusion tube. Such a configuration provides a reliable, substantially fixed angle of supply to an eye of the patient without unwanted misdirection of infusion flow, obstruction of infusion flow, kinking, twisting, etc. of the infusion tube.

[0054] To better illustrate the method and apparatuses disclosed herein, a non-limiting list of embodiments is provided here:

[0055] Example 1 includes an ophthalmic infusion support. The ophthalmic infusion support includes a base unit having a bottom surface and a top surface, a fixation device coupled to the bottom surface of the base unit to attach to a patient, and an infusion tube holding device coupled to the top surface of the base unit, wherein the infusion tube holding device is configured to hold a loop of an infusion tube an angle projecting upward from the patient.

[0056] Example 2 includes the ophthalmic infusion support of example 1, wherein the fixation device includes adhesive.

[0057] Example 3 includes the ophthalmic infusion support of any one of examples 1-2, wherein the fixation device includes a peel off backing covering the adhesive.

[0058] Example 4 includes the ophthalmic infusion support of any one of examples 1-3, wherein the infusion holding device includes adhesive.

[0059] Example 5 includes the ophthalmic infusion support of any one of examples 1-4, wherein the infusion holding device includes a peel off backing covering the adhesive.

[0060] Example 6 includes the ophthalmic infusion support of any one of examples 1-5, wherein the bottom surface is curved to match a patient's face contour.

[0061] Example 7 includes the ophthalmic infusion support of any one of examples 1-6, wherein the infusion tube holding device includes an angled top surface projecting away from the patient.

[0062] Example 8 includes the ophthalmic infusion support of any one of examples 1-7, wherein the infusion tube holding device includes a pair of infusion tube holding devices.

[0063] Example 9 includes the ophthalmic infusion support of any one of examples 1-8, wherein the infusion tube holding device includes at least one integrally formed infusion tube holding device.

[0064] Example 10 includes the ophthalmic infusion support of any one of examples 1-9, wherein the infusion tube holding device is formed from a resilient material.

[0065] Example 11 includes an ophthalmic infusion support. The ophthalmic infusion support includes a base unit having a bottom surface and a top surface, a fixation device coupled to the bottom surface of the base unit to attach to a patient, and a pair of infusion tube holding devices coupled to the top surface of the base unit, wherein the pair of infusion tube holding devices are configured to hold a portion of an infusion tube in a loop that projects upward from the patient.

[0066] Example 12 includes the ophthalmic infusion support of example 11, wherein the fixation device includes adhesive.

[0067] Example 13 includes the ophthalmic infusion support of any one of examples 11-12, wherein the fixation device includes a peel off backing covering the adhesive.

[0068] Example 14 includes the ophthalmic infusion support of any one of examples 11-13, wherein the pair of infusion tube holding devices include resilient clips.

[0069] Example 15 includes the ophthalmic infusion support of any one of examples 11-14, wherein the pair of infusion tube holding devices are integrally formed with the base unit.

[0070] Example 16 includes the ophthalmic infusion support of any one of examples 11-15, wherein the pair of infusion tube holding devices are integrally formed with the base unit.

[0071] Example 17 is a method that includes inserting a cannula into a patient's eye, coupling an infusion tube to the cannula, forming a loop with the infusion tube that projects upward from the patient at an angle, fixing an ophthalmic infusion support to the patient, and coupling the infusion tube to an ophthalmic infusion support that holds the infusion tube at the angle.

[0072] Example 18 includes the method of example 17, wherein fixing an ophthalmic infusion support to the patient includes adhering an ophthalmic infusion support to the patient.

[0073] Example 19 includes the method of any one of examples 17-18, wherein fixing an ophthalmic infusion support to the patient includes peeling an adhesive backing off to expose an adhesive, and placing the adhesive in contact with the patient.

[0074] Example 20 includes the method of any one of examples 17-19, wherein placing the adhesive in contact with the patient includes placing the adhesive directly in contact with skin of the patient.

[0075] Example 21 includes the method of any one of examples 17-20, wherein placing the adhesive in contact with the patient includes placing the adhesive in contact with a surgical drape near an eye of the patient.

[0076] Example 22 includes an ophthalmic infusion support, including a base unit, a fixation device coupled to a bottom surface of the base unit to attach to a patient, and an infusion tube holding device coupled to the base unit at an angle, wherein the infusion tube holding device is configured to hold a loop of an infusion tube at an angle projecting upward from the patient.

[0077] Example 23 includes the ophthalmic infusion support of example 22, further including a

second infusion tube holding device.

[0078] Example 24 includes the ophthalmic infusion support of any one of examples 22-23, wherein the ophthalmic infusion support is formed at least partially from a resilient material.

[0079] Example 25 includes the ophthalmic infusion support of any one of examples 22-24, wherein the ophthalmic infusion support is formed from a rigid material.

[0080] These and other examples and features of the present infusion devices, and related methods will be set forth in part in the above detailed description. This overview is intended to provide non-limiting examples of the present subject matter—it is not intended to provide an exclusive or exhaustive explanation.

[0081] The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments in which the invention can be practiced. These embodiments are also referred to herein as “examples.” Such examples can include elements in addition to those shown or described. However, the present inventors also contemplate examples in which only those elements shown or described are provided. Moreover, the present inventors also contemplate examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

[0082] In this document, the terms “a” or “an” are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of “at least one” or “one or more.” In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated. In this document, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Also, in the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, composition, formulation, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.

[0083] The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with each other. Other embodiments can be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72(b), to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features may be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter may lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments can be combined with each other in various combinations or permutations. The scope of the invention should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. (canceled)
2. A method, comprising: inserting a cannula into a patient's eye; coupling an infusion tube to the cannula; fixing an ophthalmic infusion support to the patient; selecting one or more of a pair of resilient openings in an integrally formed portion of the ophthalmic infusion support; deforming

one or more of the pair of resilient openings; and inserting the infusion tube into selected one or more of the pair of resilient openings.

3. The method of claim 2, wherein the infusion tube into selected one or more of the pair of resilient openings includes inserting at an orientation where the infusion tube projects toward the patient at an angle towards the cannula.

4. The method of claim 2, wherein fixing the ophthalmic infusion support to the patient includes pressing an adhesive.

5. The method of claim 4, wherein fixing the ophthalmic infusion support to the patient includes peeling off a backing to expose the adhesive before pressing the adhesive.

6. The method of claim 2, wherein deforming one or more of the pair of resilient openings includes deforming an integrally formed polymer portion.

7. The method of claim 2, wherein deforming one or more of the pair of resilient openings includes deforming an integrally formed foamed polymer portion.

8. A method, comprising: inserting a cannula into a patient's eye; coupling an infusion tube to the cannula; fixing an ophthalmic infusion support to a surgical drape adjacent to the patient's eye; selecting one or more of a pair of resilient openings in an integrally formed portion of the ophthalmic infusion support; deforming one or more of the pair of resilient openings; and inserting the infusion tube into selected one or more of the pair of resilient openings.

9. The method of claim 8, wherein the infusion tube into selected one or more of the pair of resilient openings includes inserting at an orientation where the infusion tube projects toward the patient at an angle towards the cannula.

10. The method of claim 8, wherein inserting the infusion tube into selected one or more of the pair of resilient openings includes forming a loop with the infusion tube and inserting two portions of the loop into the pair of resilient openings.

11. The method of claim 10, wherein inserting two portions of the loop into the pair of resilient openings includes inserting at an orientation where the loop of the infusion tube projects upward from the patient at an angle.

12. The method of claim 8, wherein fixing the ophthalmic infusion support to the patient includes pressing an adhesive.

13. The method of claim 12, wherein fixing the ophthalmic infusion support to the patient includes peeling off a backing to expose the adhesive before pressing the adhesive.

14. The method of claim 8, wherein deforming one or more of the pair of resilient openings includes deforming an integrally formed polymer portion.

15. The method of claim 8, wherein deforming one or more of the pair of resilient openings includes deforming an integrally formed foamed polymer portion.
